CHAPTER 1

Why is the BPS highly recommended?

1 Eeasy-to-learn denture fabrication system!

It has been said that many years of clinical experience are required to fabricate complete dentures that are satisfying to patients. Beginners stand little to no chance against experienced dentists when dentures are made using the traditional compound impression technique. Subtleties in the magnitude and direction of force applied by the operator in retracting the lips and cheeks during compound impression can be acquired only through experience. In retrospect, more straightforward skills and techniques that could be readily learned and passed onto younger practitioners should have been offered as part of dental education [1].

Times are changing! There is no doubt that the younger generation of dentists whose shoulders carry the future of dentistry is seeking an easy-to-learn denture fabrication system that ensures excellent results. We need a system comprised of preliminary and final impression procedures that are not technique sensitive and include a bite-registration method with a well-defined protocol. To achieve this objective, we need a standardized and systematic denture fabrication system. Without such a system, complete denture treatment would become increasingly difficult and outdated, causing younger (novice, beginner, less experienced) dentists to turn their back to complete dentures.

In this regard, the Bio-Functional Prosthetic System (BPS) is the only denture fabrication system that will fulfill the needs of those who want to improve their clinical skills and become "good (proficient) dentists" as quickly as possible [2].

2 Inviolable (fundamental) rules for success in complete denture treatment

Success in complete denture treatment is defined as follows,

A maxillary complete denture that doesn't dislodge
A suction-effective mandibular complete denture

Top priority is given to prevent the dislodgement of the maxillary complete denture. It is essential to "identify the optimum vertical dimension and horizontal mandibular jaw position and to take suction-effective impression under stable occlusion" to achieve those two objectives. Patient satisfaction is obtained as a result. Even if a great suction-effective impression is taken, the patient will have a hard time to chew if the denture moves on the ridge due to incorrect intermaxillary record. On the other hand, the denture would function well enough even without suction-effect as long as the occlusal position is stable(**Fig 1-1**) [3].

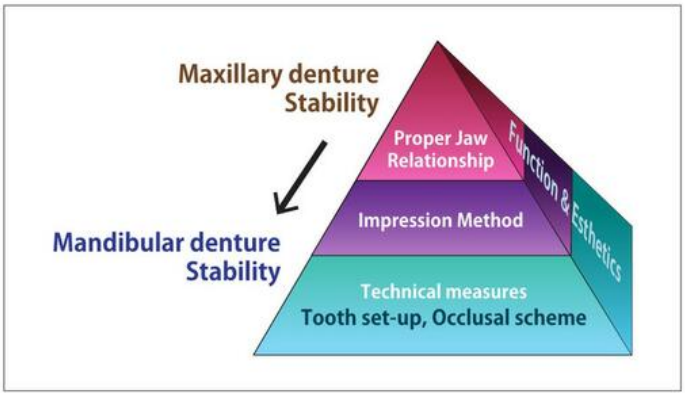


Fig 1-1 Hierarchy of patient satisfaction with complete dentures.

3 The reason why BPS is highly recommended in so many ways

When considering the intermaxillary record as the most critical step to satisfy with our complete dentures, the BPS system offers predictable results because it has a unique, 3-step process of determining jaw relationships (**Fig 1-2**).

1st phase : Primary bite registration with a Centric Tray.

2nd phase : Reconfirmation of occlusal vertical dimension with phonetics/speech-check following the final impressions.

3rd phase : Final determination of horizontal mandibular position with Gothic arch tracing using the final impressions.

Another advantage of BPS is that it allows sensory deprogramming of an improper mandibular position created by old dentures that the patient has been wearing. One hour precision impression time enables patients to forget their habitual mandibular jaw position; instead, we are able to acquire a stable mandibular jaw position in the bite registration phase just following the precision impressions.

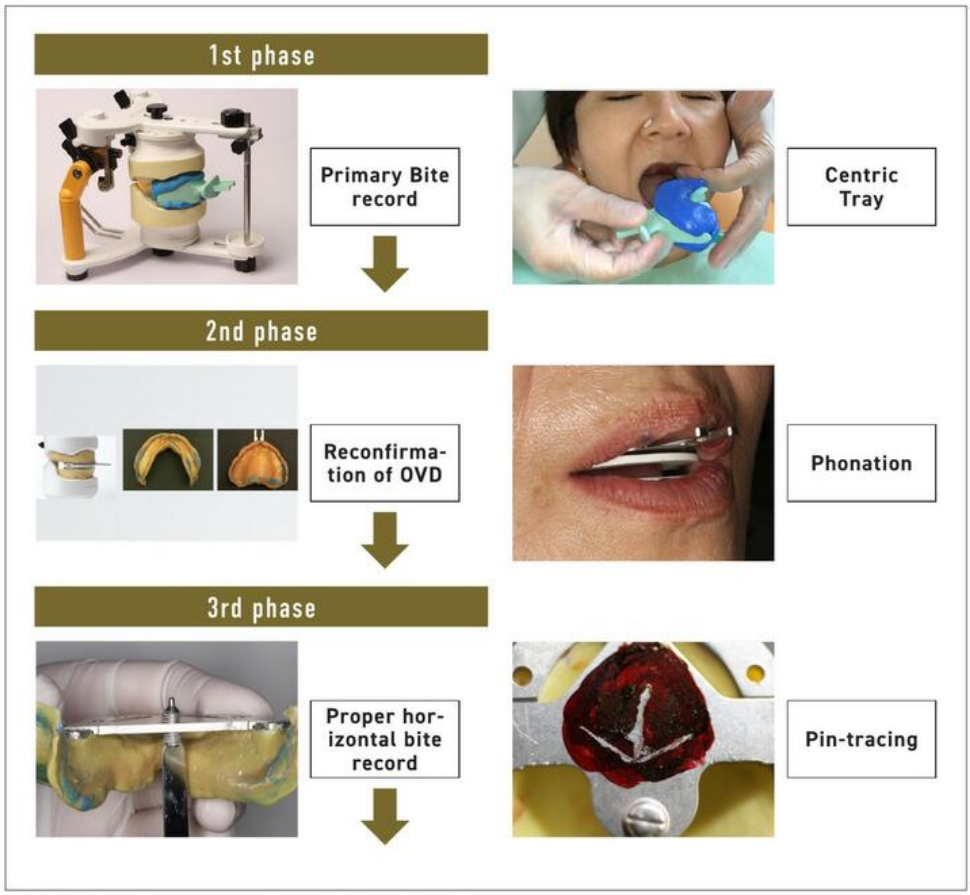


Fig 1-2 A unique, 3-step process of determining jaw relationship.

4 Denture fabrication process with BPS

The Bio-Functional Prosthetic System (BPS) originates from studies by Dr. Rainer Strack of the University of Tübingen in Germany in 1955.

It can be defined as "a system consisting of a series of steps from impression-taking, bite registration, set-up of denture teeth, polymerization and finishing to delivery that is carried out according to clearly defined BPS procedures, using Ivoclar Vivadent's devices and materials, with the ultimate goal of a happy patient." This system is currently categorized into an analog and digital workflow.

The following devices and materials are an integral part of the system when fabricating analogue BPS dentures. Regarding digital denture lab work, details will be described in the "advanced manual for the fabrication of Suction Effective Complete Denture Using BPS" (**Fig 1-3**).

1) Accu-Dent XD System for preliminary impressions.
2) Centric Tray for primary bite registration.
3) Virtual, a silicon impression material for final impressions.
4) Gnathometer M and Universal Transfer-Bow System for final bite registration.
5) Ivoclar Vivadent denture teeth (Orthosit SPE, Vivadent SPE, SR Phonares, etc.) and Stratos 100, 200 or 300 articulators.
6) IvoBase polymerization system

Dental stone is chosen according to the amount of shrinkage of the resin.

The IvoBase System is an auto curing system which does not require large scale equipment. Ideal resin polymerization is thus carried out at a programmed temperature.

The author recommends that beginners with little experience start by preparing the necessary devices and materials and follow the BPS steps precisely because there is no other system that is as simple and systematic as the Bio-Functional Prosthetic System (BPS). This way, they can first develop skills with a success rate of 70 to 80% of the time and with more experience, reach higher levels of success.



Fig 1-3 BPS product overview.

5 The function of BPS and increasing esthetic demands

The demand for quality dentures that are not only functional, but esthetic has been growing with the increasing number of lively edentulous individuals called "Active Seniors." Since the beginning of the new millennium, people have begun to enjoy life to the fullest with their individuality and personal style.

The addition of "denture coloring" to the basic BPS technique makes a denture look more personal, which also enhances the patient's individuality and becomes an integral part of the body to perform a crucial function (**Figs 1-4,5**) [4,5].



Fig 1-4 Basic BPS dentures designed to restore oral function without gingiva characterization.

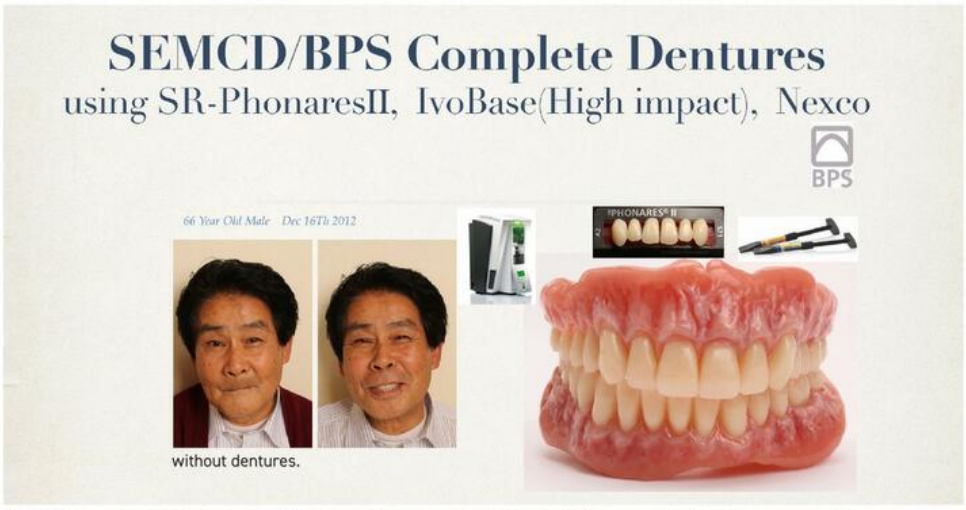


Fig 1-5 Personalized BPS dentures with enhanced function and esthetics (SR-Phonares teeth SR-Nexco were used).

6 The BPS underpinning knowledge and experience are paramount to understand SEMCD

One of the most challenging results to achieve in clinical dentistry is the suction of the mandibular complete denture. A technique for achieving effective suction of the mandibular complete denture, newly developed by the author, has become an indispensable part of the armamentarium available for current prosthodontic treatment.

It seems that the ideal way to achieve suction of the mandibular complete denture is to combine this technique with BPS. The incorporation of the concept of suction mechanism into the systematic BPS denture fabrication process will ensure a high success rate in attaining mandibular denture suction. For this objective, it is recommended that the fundamentals of BPS be learned first before moving onto the more advanced stage of acquiring knowledge and skills of mandibular denture suction and combining it with BPS [6,7].

7 *A la carte* of clinical applications of BPS

BPS offers a wide range of applications from partial, single (one-jaw) and full dentures to overdentures on natural teeth and implant-supported overdentures (**Fig1-6**). The two most significant causes of denture fabrication failure are the separate steps of impression-taking and bite registration, as well as the lack of process to set an ideal occlusal plane specific to each type of prostheses.

Denture treatment should be personalized as much as possible within patients' economic constraints since each mouth is different. In this regard, BPS is an extremely rational system that will lead to success in denture treatment.

Clinical applications of BPS

1.U & L complete dentures



2. Maximally single denture and Mandibular conical crown restoration





3. I.O.D





4. Over-dentures





5. Duplicated denture Courtesy by Ivoclar Vivadent



Fig1-6